

# Harshit Mani Tripathi

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## Summary

DevOps / Platform Engineer with production experience at Unstop, launched a user-facing cloud-native coding workspace platform. Recommended by Sr. Director of Tech and senior engineers at Unstop. Building multi-region hybrid homelab infrastructure since high-school.

## Experience

**DevOps Intern** | Unstop, Delhi

March 2026 - June 2026

- Engineered a production-grade, user-facing EKS-based coding workspace platform with zero-downtime rollouts via Kubernetes rolling updates, serving public users at scale.
- Implemented a FastAPI microservice enabling isolated, persistent coding workspaces in a multi-tenant architecture across 3 access tiers (organizer/student/evaluator), eliminating per-tenant EC2 provisioning; designed and iterated API contracts in collaboration with the backend team.
- Reduced coding workspace EBS costs by ~80% via snapshot-based images; identified and proposed ~30-40% additional savings across AWS environment through compute right-sizing and resource cleanup.
- Implemented variable-driven multi-environment Terraform configurations provisioning production-grade EKS clusters with Karpenter for dynamic node autoscaling, securing public-facing infrastructure with Traefik ingress and TLS termination.
- Built custom SigNoz log preprocessors and developer-facing telemetry dashboards; authored VPA resource-tuning scripts to optimize cluster costs.

## Infrastructure Projects

**Hybrid Infrastructure Lab**

[github.com/hudater/homelab/](https://github.com/hudater/homelab/)

- Designs and operates a cost-optimized, multi-cloud hybrid infrastructure spanning on-premise Kubernetes and OCI cloud edge workloads with cross-region high availability.
- Enforces zero-trust network isolation via OPNsense, NetBird, and VLAN segmentation across all workloads.
- Manages full lifecycle deployment via OpenTofu across OCI, k3s, and Cloudflare; infrastructure and services as code.

**Recursive DNS & Observability Stack**

[blog.hudater.dev/posts/dns-infra/](https://blog.hudater.dev/posts/dns-infra/)

- Operates core recursive DNS routing via Technitium and Unbound, handling ~500k daily queries with 75% cache hit rate.
- Runs Prometheus, Grafana, Loki, and Tempo for unified metrics, logs, and distributed tracing across hybrid workloads.

## Open Source, Community & Publications

- Linkme Generator** | [github.com/ironicbadger/linkme/pulls?q=author:hudater](https://github.com/ironicbadger/linkme/pulls?q=author:hudater)  
Contributed 5+ features in Go and automated multi-stage CI/CD deployments via GitHub Actions.
- CNCG Noida SRE Games** | [showcase.hudater.dev/sre\\_games.pdf](https://showcase.hudater.dev/sre_games.pdf)  
Investigated and resolved simulated Kubernetes microservice outages under time constraints using distributed tracing.
- Technical Writing** | [blog.hudater.dev](https://blog.hudater.dev)  
Authored in-depth deep-dives on DNS internals, core networking protocols, and Git object model.

## Skills

- Cloud & Infrastructure:** AWS, Oracle Cloud Infrastructure (OCI), Proxmox, Linux, High Availability, Cloud-Native Infrastructure
- Containers & Orchestration:** Kubernetes, Docker, Helm, Karpenter
- IaC, CI/CD & Version Control:** Terraform (OpenTofu), Ansible, Git, GitHub Actions
- Languages & Frameworks:** Go, Python, FastAPI, Bash
- Networking & Security:** OPNsense, NetBird, WireGuard, VPC, VLAN, Traefik, Keycloak, Authentik
- Observability:** SigNoz, Prometheus, Grafana, Loki, Tempo, Goldilocks (VPA)

## Education

**B. Tech CSE (Data Science)** | Noida International University  
8/10 GPA

Aug 2024 - Aug 2028